

# ShadeMapNav: six features for Google Maps and Geo GenAI roles

Research and implementation recommendations • September 8, 2026

Prepared for the ShadeMapNav author, targeting the two supplied Google SWE descriptions. Repository examined: main at `f61371c266080e02b0eaba5a243030da96cfe8f8`. This report recommends work; it does not claim the proposed features have been built.

## The recommendation

**Build a navigation system that can see why its map is wrong, improve its spatial model, and use that evidence to produce and repair a feasible outdoor plan.** Your strongest portfolio story is the connection from visual observation to spatial inference to a decision someone can inspect.

Your instinct about vision is well supported by the role descriptions. I would put vision in the map data and decision pipeline as well as the interface. An impressive demonstration would be: a building-only model misses a tree canopy; reviewed imagery changes the estimated shade; the planner changes its route; a visual scout finds uncertain access at the destination; the itinerary repairs itself while preserving the user's constraints.

These are six substantial feature recommendations, ranked by overall fit to the two roles. The ranking is my judgment, not a measured hiring probability or a claim about Google's private interview rubric.

| Rank | Feature                                           | What makes it worth interviewing you about                                                                            |
|------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 1    | Visual Shade Field                                | Segmentation and visual observations fused with geometry, an owned learning problem, and measured physical accuracy.  |
| 2    | Sun Budget with time-dependent planning           | Spatial algorithms, constrained optimization, explicit approximation error, and fast interactive execution.           |
| 3    | Visual Route Scout and Living Itinerary           | An agent selects evidence, audits a walk, executes validated tools, and repairs persistent application state.         |
| 4    | Active Survey Planner and Map Change Investigator | Data acquisition becomes an optimization problem; the system improves under a measured annotation and walking budget. |
| 5    | Find the Light                                    | Multimodal retrieval finds a specific seat or viewpoint and verifies the relevant arrival and dwell window.           |
| 6    | Shade Design Studio                               | Inverse spatial design: choose shade interventions under a budget, with an integer formulation and an LP bound.       |

**For the Geo role, lead with Features 1 and 4. For the Maps role, lead with Features 2 and 6. Feature 3 connects either specialization into a compelling GenAI product.** Features 1–3 form my preferred integrated capstone; the remaining features are meaningful expansions or alternative demonstrations, not six simultaneous commitments.

The recommended baseline respects the roadmap's free, local-first architecture: React/TypeScript/MapLibre for the product, a worker for spatial computation, offline Python for training and evaluation, and versioned derived artifacts for browser use. Keep the current text-agent provider initially. Live vision inference is a separate, profiled extension; changing providers is not the technical achievement.

## What the evidence says about these roles

The local job files match official pages retrieved during this research. Both pages displayed the role and an Apply action on September 8, 2026; neither showed a posting date. That establishes visible page state, not internal headcount.

The **AI/ML GenAI, Geo** description names ML infrastructure and GenAI experience, with preferred expertise in segmentation, multimodal fusion, spatiotemporal analysis, and shipping complex ML systems. The **Generative AI, Maps** description adds computational geometry and spatial data structures as a minimum qualification, and optimization, linear programming, and convex optimization as preferences. Both also prefer accessible technologies. These are the strongest direct signals for selecting work. [Google Careers: AI/ML GenAI, Geo](#); [Google Careers: Generative AI, Maps](#).

Public Google work supports three specific architectural choices:

- **Imagery can improve the underlying map.** Google describes detecting speed-limit signs, associating them geographically, investigating possible changes, and corroborating updates using multiple sources and operations review. That is a useful precedent for your observation-to-map pipeline; it is an internal workflow, not a public extraction API. [Google Maps, October 16, 2023: How AI and imagery keep speed limits updated](#).
- **Learned estimates and deterministic planning belong together.** Google's routing research describes evaluating learned road rewards offline, storing the results, and using fast online graph search. This supports keeping expensive inference away from every route query. It does not require reproducing Google's scale or inverse reinforcement learning. [Google Research, September 12, 2023: World scale inverse reinforcement learning in Google Maps](#).
- **Visual agents have a relevant research direction.** StreetReaderAI is a proof of concept for accessible, context-aware streetscape exploration. Its October 2025 report explicitly identifies autonomous visual exploration and full-route rehearsal as future work. That makes a measured pedestrian route scout a well-matched project direction, without proving it remains globally unique. [Google Research, October 29, 2025: StreetReaderAI](#).

Two corrections matter when presenting the project. Google's March 12, 2026 announcement already includes Ask Maps, route-tradeoff explanations, imagery-derived road details, and entrance guidance. A conversational map or landmark description therefore needs a more specific contribution. Those consumer announcements do not establish public API access to their internals. [Google Maps: Ask Maps and Immersive Navigation](#).

Also, the roadmap's claim that nobody advances the sun along a route is too strong. Fujiwara and colleagues' 2024 paper evaluates accumulated irradiance during walking using departure times, speed, and position-specific timestamps. Its example compares three predefined routes; that is different from your proposed constrained graph search and agent repair. Present your engineering contribution precisely rather than claiming the underlying idea is unprecedented. [Fujiwara et al., Building and Environment, September 13, 2024, section 6.2](#).

My inference about reviewers is straightforward: a recruiter needs to recognize relevant skills and a finished result; a hiring manager needs evidence of ownership and judgment; another SWE needs algorithms, failure cases, reproducibility, and clear interfaces. Give all three an inspectable demonstration. Public role descriptions cannot tell us which feature a particular individual will like most.

# What the current repository actually provides

The repository already supplies meaningful foundations. The implementation plans below extend these rather than replacing the app.

| Current evidence at the inspected commit                                                                                                                               | Consequence for the recommendations                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| app/lib/shade/ShadeField.ts:84 defines geometry-backed point, edge, coverage and time-sweep queries. PrismProvider at line 104 accepts tile and Overpass sources.      | Add visual evidence through a deliberate field/composition interface; the reserved canopy source name is not an implemented canopy model.                |
| app/hooks/useNavigation.ts:1154 samples an edge batch at one selected time. app/lib/routing.ts:580 has distance/shaded-distance labels and caps labels per node.       | Traversal-time exposure and proof of approximation quality are new work. The present bounded Pareto search is not an established exact solver.           |
| app/lib/agent/llmClient.ts:25 supports text and tool messages. Seven tools are declared in tools.ts:113.                                                               | Image transport, visual model evaluation and evidence-aware tools are new capabilities.                                                                  |
| tools.ts:448 already accepts intermediate stops, but returns after starting route calculation. useNavigation.ts:979 already uses generation counters and cancellation. | Expose terminal results and versioned plans; reuse existing stale-result protection. Do not rebuild multi-stop support or claim cancellation is missing. |
| Eighteen scripted agent scenarios are registered in agent/__tests__/scenarios/index.ts:8; model and tools are mocked in the harness.                                   | Extend C1 into real-executor and separately labeled live-model evaluations. Replay passing is not measured model competence.                             |
| ShadeField.ts:193 uses hand-set source confidence priors. The agreement harness compares two methods using synthetic geometry.                                         | Calibration against independent physical observations remains missing. Model agreement cannot establish physical accuracy.                               |
| OSM mode/access tags enter at overpass.ts:220 but are lost rebuilding sidewalk edges in routing.ts:393. Geographic time uses longitude rounding in timezone.ts:7.      | Preserve access metadata and fix IANA/date-specific local time before claiming access constraints or reliable temporal plans.                            |

The audit ran a targeted check: `npm test -- app/lib/agent/__tests__/agentScenarios.test.ts app/lib/shade/__tests__/agreement/agreement.test.ts app/lib/__tests__/routing.test.ts`. All **110 tests across three files passed**. The agreement run reproduced 150 cases, mean 2.6 percentage points, worst 62.5 points, and severe disagreements in 3.3% of sidewalk readings. Each case supplies left/right readings. These are synthetic method-agreement results, not real-world accuracy.

This was a source audit with selected tests, not a fresh browser audit, full CI run, live-agent benchmark, or production performance study. The earlier shadow-index speedup is a synthetic Node microbenchmark; it should not become a claim that entire browser routes became thousands of times faster.

# 1. Visual Shade Field

**The experience.** Select a sidewalk sample and inspect the original photo, its sky/canopy/building masks, the estimated solar obstruction, and the route consequence. A tree-lined segment that the building-only model treats as exposed can become a useful alternative when supported by observations. The evidence has a capture date and a visible coverage limit.

**Why I chose it.** This is the closest direct match to the Geo description's segmentation, fusion, spatiotemporal analysis and ML lifecycle requirements. It also forces the geometric reasoning required by the Maps role. Its distinctive value is a measured change in a spatial decision.

**Implementation.** Start with one walkable neighborhood and calibrated panoramas at known sidewalk positions. Use your own photos or imagery whose permissions cover the intended processing and distribution. Record position accuracy, camera orientation, camera height, capture time, season, image hash, and annotation version. Reject inadequate pose estimates instead of attaching precise coordinates to uncertain evidence.

Use SAM 2.1 to assist reviewed mask creation, with people supplying the semantic class labels. SAM provides prompted masks, not automatic ground truth, semantic taxonomy, or depth. Its official code and checkpoints are available under Apache 2.0; published accelerator performance is not a browser benchmark. Train a compact supervised segmentation head or small segmenter on reviewed labels and retain a frozen-model baseline. [Meta FAIR: SAM 2 official implementation and training documentation, releases 2024.](#)

At each observation position, construct an angular field for sky visibility and occlusion. Query it using solar azimuth and elevation. Distinguish opaque buildings from partly transmitting canopy, and keep the result's units explicit. Panorama-based segmentation and canopy transmission have an original research precedent, including measured irradiance validation; the paper also identifies errors around reflective buildings behind foliage. [Fujiwara et al., 2024: A panorama-based technique to estimate sky view factor and solar irradiance considering transmittance of tree canopies.](#)

**A panorama describes visibility from its capture point. It does not create a 3D canopy model covering the whole street.** My proposed first release limits interpolation to distances validated with withheld observations. Broader coverage can later use surveyed multi-view geometry or suitable lidar. Normalize coordinate systems, vertical units and acquisition dates before fusion. Lidar includes surface features that bare-earth DEMs remove. [USGS: Lidar data versus a digital elevation model.](#)

Compare geometry-only, image-only and fused predictions. Fit a logistic or small tree-based residual correction only if independently labeled observations justify it. Keep current heuristic confidence separate from a learned probability and from a canopy transmission fraction. Extend the field boundary with observation coverage, source IDs, model version and validity period; do not force an angular observation into the existing building-prism type.

**Proof and stopping rule.** Publish class IoU/boundary error, observed direct-sun classification, false-shade rate, calibration and route-level exposure error. Split by street/neighborhood and capture day; keep calibration and final test data separate. Include pose perturbations, missing imagery and leaf-season changes. Publish paired uncertainty intervals using location/walk clusters. Ship the learned component only when held-out decision quality improves without an unacceptable false-shade regression.

**First reviewable slice.** A small, honestly bounded corridor with one replay showing how a photo becomes masks, changes a field estimate and affects the route, plus a dataset card, ablation report and model/data rollback. It is an ambitious ML feature; collecting repeat observations is likely harder than integrating the model.

## 2. Sun Budget with time-dependent planning

**The experience.** “I have 70 minutes. Find somewhere shaded to read for 20, keep estimated direct sun under eight minutes, and bring me back.” The app shows feasible destinations, route choices and the constraint that limits each alternative. Moving departure time changes the whole plan.

**Why I chose it.** This is the strongest immediate fit to the Maps role's spatial algorithms and optimization requirements. It turns the existing shadow engine into a decision system and gives every subsequent AI feature a deterministic tool worth calling. Keep it central even when adding vision.

**Implementation.** Finish the roadmap's temporal/performance foundations: correct geographic timezones, reliable geometry coverage, a route benchmark, and worker execution. Extend `ShadeField.sweep` and Track H rather than introducing a second solar model. Cache by geometry/observation version, region and time cell; measure time discretization error at shadow boundaries.

Use a time-expanded reference graph. A state carries position, time, used exposure and necessary visit state. Movement, waiting and dwell are explicit actions. Integrate exposure over the travel interval, rather than pricing an edge only at departure. For a binary direct-sun definition, the modeled resource is:

```
sun_minutes = sum over intervals(duration_minutes × direct_sun_fraction)
```

A continuous canopy-transmission model gives a different quantity—equivalent full-sun time—and must be labeled separately. Neither is a medical safety guarantee. Deadlines, maximum detour, required stop duration and access exclusions are hard constraints. Preferences rank feasible plans.

Start interactive planning with bounded label-setting search and publish the approximation. Earlier arrival does not automatically dominate later arrival: reproducing the later state's time may require waiting in the sun or at an unavailable place. State precisely which pruning rule is valid under your waiting and resource model. Preserve OSM access tags through the sidewalk transformation; do not infer a legal crossing solely from parallel graph edges.

For correctness, implement exhaustive enumeration on tiny time-expanded fixtures with a finite horizon and positive-duration actions. Add an independent binary-flow or scheduling formulation offline where useful. Compare against static-time routing and departure-grid search. A solver's time limit cannot establish infeasibility: OR-Tools explicitly distinguishes optimal, feasible, infeasible and unknown states. [Google OR-Tools: CP-SAT solver documentation](#).

Return a structured result containing status, route/plan version, satisfied constraints, estimated exposure, search limits, and any bound. Proven infeasibility applies only to the modeled discretized instance; incomplete data or a capped heuristic may justify only “no plan found within this search.”

**Proof and stopping rule.** Publish hard-constraint violations, oracle objective gap, discretization error, expanded states, memory, and cold/warm p50/p95 latency on named hardware. Separately evaluate real-world exposure using Feature 1's held-out observations; agreement with a synthetic oracle verifies the algorithm, not the world's geometry. Keep speed/accuracy curves, including failures, rather than one favorable example.

**First reviewable slice.** Track H1–H5 for a bounded area, with a route that looks good at departure but loses when the sun advances, a dwell/return case, and an impossible-budget case. An optional LP relaxation supplies a lower bound for a minimization model; it does not turn the discrete itinerary problem into convex optimization.

### 3. Visual Route Scout and Living Itinerary

**The experience.** “Inspect the walk to this reading spot. Check the entrance and the uncertain section, then keep me within my sun budget.” The agent chooses useful views, produces source-linked observations and unknowns, and commits a validated itinerary. “I’m 15 minutes late” or “that entrance is blocked” creates a visible revision that preserves unaffected stops.

**Why I chose it.** This is the clearest visual-agent demonstration for these roles: perception, selective information gathering, tool use, persistent state and recovery all affect an actual route. It connects to StreetReaderAI’s proposed full-route rehearsal while giving your project its own shade-specific objectives and evaluations. The original is explicitly a research prototype, not evidence that the entire proposed workflow is already a developer service. [Google Research, October 29, 2025: StreetReaderAI](#).

**Implementation.** Create a versioned `Trip` with stop/entrance IDs, time windows, dwell, constraints, selected route legs and evidence references. First fix C4/C11 and E5: await terminal planner results, reuse generation/cancellation protection, reject stale writes, and show a plan diff. Include an expected-version check and idempotency key when committing a revision.

Add bounded tools such as `get_route_evidence`, `inspect_view`, `query_shade`, `solve_trip`, `validate_trip` and `commit_revision`. The agent chooses which candidate views to inspect next under an explicit inference/time budget. Each visual assertion must carry a region or mask, image ID, capture time, candidate edge/entrance association and confidence or an unknown result. A separate validator checks schema, graph references, evidence freshness and plan constraints.

Initially, inspect a curated image corpus processed offline. The existing text agent can select observations and request comparisons without adding a paid endpoint. To demonstrate live image-conditioned behavior, add a local VLM adapter and image messages as a later measurable step. A current candidate is Gemma 4: Google’s model card documents image understanding and structured tool use, and identifies an Apache 2.0 license. Suitability and quantized performance on your equipment still require testing. [Google DeepMind: Gemma 4 model card, updated July 30, 2026](#).

Treat OCR, place text and model output as data with no authority to change tool permissions. Reject unsupported facts even when their wording is persuasive. A photograph can show an apparent obstacle at capture time; it cannot guarantee current passage, certify an accessible route, or authorize a crossing. Start with route preparation and user-confirmed replanning. Include keyboard operation, screen-reader summaries and a text equivalent for each visual finding.

**Proof and stopping rule.** Compare the scout with fixed-interval image sampling and a single-pass route summary under equal image/token budgets. Score verified issue discovery, false claims, evidence association, uninspected coverage, valid completed plans, stale-write rejection and recovery. Test missing views, conflicting dates, prompt injection in signs, unsupported access claims, timeouts and budget exhaustion. Evaluate actual final trip/map state; keep model correctness distinct from deterministic replay results.

**First reviewable slice.** One complete route rehearsal and repair, with an inspectable trace showing why the agent requested a second view and exactly what changed in the trip. A cached-evidence demo should say so; do not imply it inspected a new photograph live. This should precede continuous camera navigation, which needs additional location tracking and crossing-data foundations.

## 4. Active Survey Planner and Map Change Investigator

**The experience.** “I have 20 minutes to help improve this area.” The app proposes a short survey walk, explains which observations would resolve important uncertainty, and compares newly collected evidence with prior captures. A removed tree or new awning becomes a reviewed, versioned map change.

**Why I chose it.** This directly addresses your data-gathering idea and fills the largest gap in the current ML story: owning how useful data is selected, labeled, validated and deployed. It demonstrates spatiotemporal reasoning and ML infrastructure, not just consuming a pretrained model.

**Implementation.** Build on Feature 1's corpus and Feature 2's planner. Each candidate observation has position, expected collection effort, image age, geometry/vision disagreement and estimated effect on useful route choices. Score acquisition by a combination of uncertainty, spatial diversity and estimated decision impact. These scores are proposed heuristics, not guaranteed information gain. Solve a budgeted visit-selection problem with travel time, dwell for capture and return included.

Keep a random audit sample and compare several acquisition strategies. Spatial active-learning research supports considering spatial dependence, but evidence from crop mapping is not a promise of gains for street photographs. [Kaijage, Belgiu and Bijker, Sensors, March 26, 2024: Spatially Explicit Active Learning for Crop-Type Mapping from Satellite Image Time Series.](#)

For change detection, register repeat observations before comparing them. Track camera pose, seasonal foliage, illumination and temporary occluders so these are not silently treated as structural change. Begin with one change class and human adjudication. The agent can ask for a corroborating view, but a reviewed record—not its prose—activates an update. Keep changes inside the app's derived overlay; this proposal does not authorize automatic edits to external maps.

Use an append-only observation history with reviewed change events, affected field cells/graph edges and an expiry policy. Retraining reads a pinned manifest; calibration uses a separate split; evaluation compares candidate and current versions; a data/model manifest selects the active release. Rollback restores the whole compatible version. If deletions affect training inputs, record that relationship and define the retraining policy.

This resembles the useful structure of Google's published update workflow: detect a possible change, gather corroboration, then update with review. The analogy concerns engineering process, not equal data coverage or operational scale. [Google Maps, October 16, 2023: How AI and imagery keep speed limits updated.](#)

**Proof and stopping rule.** Plot held-out physical/route error against annotation minutes and collection distance. Compare random selection, spatially uniform selection, uncertainty-only selection and the proposed strategy with equal budgets. In retrospective experiments, hide labels until acquisition; label them as simulations of collection. Also measure independently confirmed change precision, localization error, review effort, and downstream routing regressions. Cluster resampling by walk/location; adjacent frames are not independent evidence.

**First reviewable slice.** An initial corpus, two or more acquisition rounds, one adjudicated change and a reproducible model/data rollback. Retain the simpler strategy if the clever selector does not improve learning efficiency. This is a high-value follow-on to a working vision baseline, not a reason to postpone collecting the first ordinary observations.

## 5. Find the Light: a specific place at the right time

**The experience.** “Find a place like this photo where I can read in shade from 4:30 to 5:00,” or “Find a west-facing viewpoint I can reach before the sun disappears behind the skyline.” The result identifies the actual seat, terrace or viewpoint, shows an authorized reference photo, and includes an achievable arrival, dwell and return plan.

**Why I chose it.** This makes your sunset idea useful while adding a concrete multimodal retrieval problem. It combines image/text understanding with spatial joins, temporal filtering and explainable ranking. It is more differentiated than retrieving a pleasant-looking café and letting the model guess the sunlight.

**Implementation.** Curate perhaps 20–50 micro-locations as an initial product scope, not an accuracy target. Store entrances, point/polygon geometry, view direction, access information, photo permissions/capture dates and any observed occlusion field. Keep verified facts separate from visual attributes and subjective preferences. A business centroid is not a patio table.

Embed authorized images and text with a pinned image-text encoder such as SigLIP 2. Google's official release provides code and checkpoints for multilingual vision-language representation. For a small corpus, exact vector similarity over a local array is sufficient; evaluate before introducing a vector service. [Google Research, 2025: SigLIP 2 official release](#).

The retrieval system produces candidates and relevance scores. Hard filters then check access, deadline and travel budget using the actual graph; the field evaluates shade over arrival and dwell. An explicit reranker balances requested visual attributes with feasible plans. The generative model extracts intent and explains the accepted evidence, while the retrieval and temporal tests remain independently inspectable. A similarity score is not evidence of opening hours, quietness or future shade.

For a sunset viewpoint, use the solar path and a measured or modeled horizon profile at that position. Distinguish sunset below the astronomical horizon from disappearance behind a building. Missing horizon geometry yields an unknown visibility interval. Clouds affect the eventual view; a predicted solar direction is not a promised photograph.

**Proof and stopping rule.** Compare tag/keyword search, image-text retrieval, and retrieval plus spatial/temporal filtering. Hold out queries and places; use blind human relevance judgments for subjective similarity and deterministic checks for constraints. Report Recall@k or nDCG, feasible-result rate, incorrect sunlight claims and route/dwell exposure error. Include visually similar but inaccessible places and photos from a different season as hard negatives. Publish a candidate rejection example as well as successes.

**First reviewable slice.** A query photo retrieves several plausible places, but only one passes the user's dwell-time shade and return constraints. Its result shows the photo, exact micro-location and why it was chosen. This is a visible product layer on Features 1–3 and a natural extension of the roadmap's Find the Light idea.

I would defer photorealistic generated future views. Google's April 2026 imagery announcement distinguishes private-preview imagery grounding for generated media from imagery analytics and experimental understanding models. Those are different capabilities and access stages. [Google Maps Platform, April 22, 2026: Three new ways to build with real-world imagery and AI](#). If you later add generative relighting, label it an illustration and keep the measured geometry overlay as the planning evidence.

## 6. Shade Design Studio

**The experience.** “Where would two shade structures most improve this walking loop between noon and 3 PM?” The app proposes candidate placements and shows the modeled change in exposed minutes. Users can inspect a placement, change the budget or priorities, and see why another solution wins.

**Why I chose it.** This upgrades the roadmap's playful Shadow Lab into a substantive inverse-design problem. It offers the clearest bounded demonstration of linear and integer optimization alongside computational geometry. It is especially well aligned with the Maps posting, and creates a visually understandable interview discussion about objective choice and approximation.

**Implementation.** Begin with a finite set of manually approved candidate locations and standardized canopy shapes. Keep receiver positions, time samples and demand weights explicit. The geometry engine precomputes whether each candidate shades each currently exposed receiver/time sample. Use union coverage so overlapping shadows are not counted twice.

For a simple proposed maximum-coverage model, let  $x[j]$  choose candidate  $j$  and  $z[k]$  represent coverage of receiver/time sample  $k$ . Let  $a[j,k]$  indicate modeled coverage,  $c[j]$  be a cost, and  $w[k]$  a nonnegative demand weight:

```
maximize sum_k w[k] * z[k]

subject to sum_j c[j] * x[j] <= budget

z[k] <= sum_j a[j,k] * x[j]; 0 <= z[k] <= 1

x[j] in {0,1}, plus explicit placement exclusions
```

This objective counts modeled weighted coverage for the chosen samples. It is not automatically the same as maximizing benefits after users reroute. Evaluate that second question by rerunning the route planner; if routing behavior becomes part of the optimization, disclose the more complex formulation and any heuristic search.

Solve small instances with exhaustive enumeration and an offline OR-Tools integer formulation. Use Boolean  $z[k]$  for CP-SAT; this is equivalent for the binary-coverage objective with nonnegative weights. Run a greedy baseline in a browser worker for interactive experiments. Relax  $x[j]$  and  $z[k]$  to  $[0,1]$  to obtain an LP upper bound for this maximization problem; the fractional result is a bound, not a buildable design. CP-SAT is an integer solver, whereas the relaxation is linear/convex. Scaling coefficients to integers introduces another approximation to document. [Google OR-Tools: CP-SAT](#); [Google OR-Tools: Linear Optimization](#), updated August 28, 2024.

The generative layer converts a user's design brief into editable constraints and explains solver-backed tradeoffs. It must not invent valid locations or certify structures. Add elevated-canopy occlusion and ground-receiver tests before supporting overhead shade structures: the current `shadowIndex.ts:227` excludes points inside building footprints, so reusing that convention would miss shade beneath a canopy. Existing building primitives can first model solid obstacles shading adjacent ground. Tree growth, partial transmission and seasonal conditions can be later scenario inputs supported by Feature 1. Show these as modeled assumptions, not validated urban cooling or structural engineering.

**Proof and stopping rule.** Publish objective value, exact/LP bound, gap, runtime and sensitivity to demand assumptions. Validate geometric coverage against independent simple fixtures and show a case where greedy selection loses. Compare modeled route exposure before/after while keeping original trips and weights available for reproduction. If no real demand dataset exists, label weights as scenarios instead of implying observed pedestrian traffic.

**First reviewable slice.** A small public-space scene where selecting the individually best shade object twice is worse than a complementary pair, demonstrated with the exact reference. This can be a self-contained Maps-role specialization after the core field and planner; it need not become a separate municipal planning product.

# How to build this without losing the roadmap

Treat this report as a role-specific refinement, not a replacement priority file. The existing roadmap remains the owner of implementation sequencing. Its strongest change for these jobs would be promoting a bounded vision/data specialization from an optional later idea into the next major capability after temporal and agent-result foundations.

| Stage                                        | Concrete exit artifact                                                                                              | Existing tracks                                                                         |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Establish trust                              | Correct time handling, intact access metadata, usable method links, pinned route benchmark and source-level claims. | D0, G2, G8, P1/P4; inspect current state rather than blindly following old status text. |
| Finish the decision and execution foundation | One time-aware trip with dwell/return, a completed-result contract and a reproducible repair.                       | A5/A6, H1–H5, C4/C11, E5. C1 replay exists; extend it.                                  |
| Build the vision capstone                    | A bounded observed corridor, reviewed labels, geometry/image/fusion comparison and one visible route improvement.   | Add a focused Reality Check brief extending A's field and P's evidence.                 |
| Demonstrate visual agency                    | The scout chooses views, reports evidence and unknowns, then repairs the same trip object.                          | C plus B's route/guidance data; start as preparation, not live crossing guidance.       |
| Choose the next emphasis                     | Active collection for Geo; inverse shade design for Maps; Find the Light for the most approachable consumer demo.   | New bounded briefs; preserve a single coherent product.                                 |

These are staged deliverables, not a promised calendar. Time depends principally on the available capture equipment, repeat observations, labeling effort and model hardware, none of which was established in this research. The initial dataset should grow from coverage and error analysis rather than a cosmetic sample-count target.

**Data and compute choices.** Use self-collected photographs for the first corpus, with explicit contributor permissions and minimized location retention. USGS describes its 3DEP point-cloud collection as public domain, and its WESM metadata exposes project acquisition dates, quality and availability. Check the selected work unit before assuming useful canopy coverage. The catalog's public-domain statement was available through first-party indexed text; the direct catalog open returned 403. [USGS: 3DEP lidar point-cloud collection, metadata 2025](#); [USGS: 3DEP Spatial Metadata](#).

Do not build the default training corpus by scraping Street View. The reviewed standard non-EEA Maps terms restrict extraction, model training/testing/validation and constructing a tree-location index from Street View. Service-specific exceptions exist, so this is not a claim that all AI use of Maps is prohibited; the current Grounding Lite exception does not grant a general training corpus. [Google Maps Platform Terms, section 3.2.3](#); [Service Specific Terms, Grounding Lite](#).

Keep raw imagery, training runs and large-model inference offline initially; deliver compact, versioned fields, reviewed observations and embeddings. Browser model inference is optional. ONNX Runtime documents WebGPU execution, but support for a particular exported model and its mobile memory/latency must be measured. [ONNX Runtime: Using the WebGPU Execution Provider](#). Free software does not make collection or computation costless; report wall time and hardware rather than inventing a zero-cost deployment claim.

Kubernetes, Spark, a vector database and an additional paid LLM are unnecessary prerequisites. A global imagery reconstruction system is beyond the proposed first scope. An intentionally small model and a trustworthy deployment/evaluation loop can satisfy the ML engineering story more convincingly than an unvalidated large model.

# What would make the finished project persuasive

Build one evaluation and debugging workbench across the features rather than counting tests as a seventh product feature. A reviewer should open a failed request and see the source photo, field/data version, model input/output, chosen tools, solver status, final application state and the scorer's explanation.

| Evaluation layer                         | Evidence it provides                                                                                     | What it does not establish                         |
|------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Existing recorded C1 scenarios           | Orchestrator contract and regression behavior for scripted model/tool responses.                         | Live-model judgment or end-to-end task completion. |
| Real executors with frozen external data | Actual planner, validator, state transitions, cancellation and reproducible failure handling.            | Changing real-world data quality.                  |
| Separate bounded live-model runs         | Held-out task success, unsupported claims, recovery, latency, tokens and cost/usage per successful task. | Universal reliability from a small task set.       |
| Independent field observations           | Physical prediction error, false-shade rate, geographic/seasonal generalization.                         | Medical outcomes or guaranteed accessibility.      |
| Solver and geometry oracles              | Correctness and approximation gaps for explicitly modeled instances.                                     | Correctness of the physical-world inputs.          |
| Named-device product tests               | Worker responsiveness, cold/warm latency, memory, fallback and accessibility of the actual flow.         | Performance on every phone or at city scale.       |

For probability claims, report reliability diagrams with sample counts and proper scoring rules. Brier loss combines calibration and discrimination, so a lower value alone cannot establish better calibration. [scikit-learn: Probability calibration, documentation accessed September 8, 2026](#). Report unknown coverage and refused/partial results alongside success; otherwise conservative behavior can appear falsely perfect.

Predeclare the dataset split, baseline, metric and release criterion before running the final evaluation. Keep tuning data separate. Show paired results and uncertainty across independent walks or places; aggregate neighboring frames as clusters. For model changes, include at least one ablation removing vision and one removing agent-selected inspection. This determines whether the costly component contributes.

**The hiring package I would prepare:** a short demo of one visual observation changing one real planning decision; an architecture diagram and design note; a reproducible benchmark command with pinned commit/model/data versions; a dataset/model card; a failure gallery; and a traceable repair example. Include keyboard and screen-reader access to the evidence. These are recommendations for making your competence legible, not an asserted Google screening checklist.

A future résumé claim could read: “Built an image-informed pedestrian planner that fuses reviewed canopy observations with geometric shade predictions; evaluated on held-out streets and dates, and integrated a tool-using agent that audits and repairs constrained itineraries.” Add numerical improvements only after those exact results exist.

**Recommendation confidence:** high that these features match the supplied technical requirements; moderate that this ranking maximizes portfolio value for an individual reviewer; unproven that any proposed model will improve this app until local data and benchmarks exist. Public Google research and product announcements inform relevance, not access to internal systems or knowledge of hiring decisions.

Research stopped after the six recommendations had primary support for consequential external claims, concrete repository integration points, independent baselines, verification criteria and bounded limitations. The main disconfirming findings—existing Google features, prior traversal-time exposure research and already implemented repository foundations—are reflected in the recommendations. No feature implementation, paid-service activation, external map edit or roadmap rewrite was performed.